

absolute deviation suggests that the effectiveness is deviating significantly from the mean, which might not be preferable as it can lead to inconsistent outcomes. FAD is a valuable metric to assess the degree of fluctuation in the effectiveness of a feature over a specific period. It is particularly useful for understanding the consistency and stability of the feature's performance.

- **Monthly absolute deviation (MAD):** This denotes a variation of the feature absolute deviation calculated at the end of every month. It quantifies the amount of variation in the monthly feature effectiveness measure (MFEM) values from the mean. MAD is instrumental in assessing the monthly variability of the feature's effectiveness. To calculate MAD, the formula involves finding the absolute difference between each monthly MFEM data point (x) and the average of the monthly MFEM values (μ). The absolute differences are summed and divided by the total number of data points (n) to obtain the monthly absolute deviation.

$$\bullet \quad MAD = \sum_{i=1}^n \square |\mu - x_i| \dot{\div} \frac{\dot{\div}}{n}$$

- μ = Average of MFEM
- x = Each monthly data point
- n = Total data points

- **Quarterly absolute deviation (QAD):** This is designed to measure the variation in the FEM over a broader timeframe, specifically at the end of every quarter. It provides a more comprehensive perspective on the feature's stability or variability by considering the quarterly FEM values over the three months of a quarter. The formula for calculating QAD is similar to MAD, involving the absolute difference between each quarterly FEM data point (x) and the average of the quarterly FEM values (μ). The absolute differences are summed and divided by the total number of data points (n) to obtain the quarterly absolute deviation.

$$\bullet \quad QAD = \sum_{i=1}^n \square |\mu - x_i| \dot{\div} \frac{\dot{\div}}{n}$$

- μ = Average of QFEM
- x = Each quarterly data point
- n = Total data points

Defining the feature constants

The growth constant (GC) and deviation constant (DC) are critical numerical parameters in the context of measuring and assessing the effectiveness of software features. These

constants play a pivotal role in the calculation of the feature effectiveness indicator (FEI) and help establish expectations for feature performance. Here's an expanded explanation of these constants.

- **Growth constant (GC)**

The growth constant, denoted as GC, is a fixed numerical value that serves as a fundamental parameter in evaluating the anticipated rate of growth or change in the context of software feature effectiveness. It is a mathematical representation of how rapidly a specific metric or quantity associated with a feature is expected to either increase or decrease over a defined period. In practical terms, GC can be thought of as a projection or expectation regarding the feature's performance improvement. For example, if a feature has a GC of 10%, this means that it is expected to exhibit a growth rate of 10% over a specified timeframe. In other words, the feature's effectiveness should ideally increase by 10% during that period. This growth rate can be a key reference point for assessing whether the feature is performing as expected and meeting the predetermined growth objectives.

- **Deviation constant (DC)**

The deviation constant, denoted as DC, is another fixed numerical value used in the context of software feature evaluation. Unlike GC, which focuses on growth or change, DC pertains to the expected rate of deviation or variation within certain boundaries. It provides insight into the tolerance for fluctuations or deviations in the effectiveness of a feature. For instance, if a feature has a DC of 10%, this indicates that its performance can reasonably deviate by up to 10% from standard or expected calculations without raising significant concerns. In essence, DC helps set acceptable limits for variations in the feature's performance.

Both GC and DC should be thoughtfully determined to align with the project's goals, and they play a crucial role in calculating the feature effectiveness indicator (FEI), which is central to measuring feature effectiveness. These constants are not arbitrary values but are derived based on the unique business context and the nature of the feature in question. For example, a project aimed at rapid growth and innovation may have a higher GC to reflect aggressive growth targets, while a project focused on stability and risk mitigation may have a lower GC and a narrower range for DC to minimise performance variations.